BCA309T: Cryptography & Network Security

ACADEMIC SOLUTIONS MANUAL

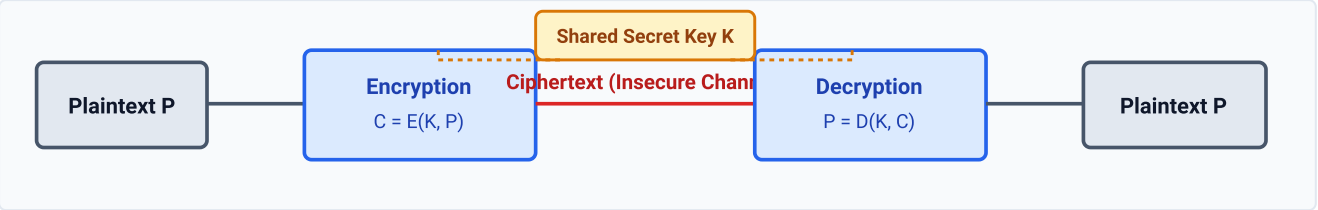
Assignment 4: Introduction to Symmetric Cryptography (Complete Solutions)

Q1 [10 Marks] Principles of Symmetric Cryptography & Kerckhoffs's Principle

Symmetric cryptography (conventional/single-key encryption) relies on a single shared secret key K for both encryption and decryption. The core components are:

- **Plaintext (P):** The original intelligible input data.
- **Secret Key (K):** Value independent of plaintext fed into algorithm; key space must prevent brute-force.
- **Encryption Algorithm (E):** Mathematical transformation producing ciphertext: C = E(K, P).
- **Ciphertext (C):** Scrambled, unintelligible output transmitted across insecure channels.
- **Decryption Algorithm (D):** Inverse mathematical transformation recovering plaintext: P = D(K, C).

Kerckhoffs's Principle (1883): States that a cryptosystem must remain completely secure even if everything about the system (including algorithms, internal state formats, and hardware implementations) is public knowledge—except the secret key. Shannon rephrased this as *"the enemy knows the system."* Security must never depend on obscurity.



Q2 [10 Marks] Stream vs. Block Ciphers, Confusion & Diffusion

Stream Ciphers encrypt continuous data bit-by-bit or byte-by-byte by generating a pseudorandom keystream k_i XORed with plaintext: $c_i = p_i \oplus k_i$ (e.g., RC4, ChaCha20). They have low latency and zero error propagation. **Block Ciphers** partition plaintext into fixed-size blocks (e.g., 64 or 128 bits) and encrypt each block as an atomic unit using identical keys (e.g., AES, DES).

Claude Shannon's Principles:

- **Confusion:** Conceals the relationship between the ciphertext and key. Implemented via non-linear substitution (S-boxes) so intercepting ciphertext yields zero clues regarding key bits.
- **Diffusion:** Spreads the statistical properties of a single plaintext bit across many ciphertext bits. If one plaintext bit flips, approximately 50% of the ciphertext bits should flip (Avalanche Effect). Implemented via permutations and linear mixing.

BLOCK CIPHER MODES OF OPERATION (Q2 CONTD.)

Mode	Operational Principle	Strengths	Weaknesses
ECB (Electronic Codebook)	Each block is encrypted independently with key K: $C_i = E_K(P_i)$.	Simple; parallel execution; bit errors stay within one block.	Identical plaintext blocks yield identical ciphertext; leaks data patterns (e.g., ECB Penguin). Insecure.
CBC (Cipher Block Chaining)	Each plaintext block is XORed with previous ciphertext block: $C_i = E_K(P_i \oplus C_{i-1})$; uses IV.	Hides plaintext patterns; randomized by IV. Standard for bulk data.	Encryption is strictly sequential (cannot parallelize); 1-bit transmission error corrupts 2 blocks.
CFB (Cipher Feedback)	Encrypts previous ciphertext block to generate keystream: $C_i = P_i \oplus E_K(C_{i-1})$.	Converts block cipher to stream cipher; processes arbitrary stream lengths without padding.	Encryption is sequential; errors in ciphertext propagate through the feedback shift register.
OFB (Output Feedback)	Successively encrypts output keystream: $O_i = E_K(O_{i-1})$; $C_i = P_i \oplus O_i$.	Keystream can be precomputed; bit errors in ciphertext do not propagate to other bits.	Vulnerable to bit-flipping attacks if integrity tag is absent; sender/receiver must maintain exact synchronization.
CTR (Counter)	Encrypts successive counter values: $C_i = P_i \oplus E_K(\text{Nonce} \| \text{Counter}_i)$.	Fully parallelizable in encryption/decryption; fast random access to arbitrary blocks; highly secure.	Counter reuse with same key completely compromises confidentiality (two-time pad vulnerability).

Q3 [10 Marks] **Historical Development of Symmetric Cryptography**

- **Classical Antiquity:** The *Caesar Cipher* used fixed monoalphabetic shifts (shift of 3). Easily broken due to invariant character frequencies.
- **Rotor Era & WWII:** The German *Enigma machine* used multi-rotor mechanical wirings and a plugboard to cycle polyalphabetic substitutions dynamically. Marian Rejewski, Alan Turing, and Gordon Welchman cracked Enigma using statistical reduction and electromechanical "Bombes."
- **Shannon's Information Theory (1949):** Claude Shannon published *"Communication Theory of Secrecy Systems,"* establishing mathematical proofs for perfect secrecy (One-Time Pad) and formalizing confusion and diffusion.
- **Feistel Architecture & DES (1970s):** Horst Feistel designed Lucifer at IBM, introducing the Feistel network. In 1977, NBS (now NIST) standardized the *Data Encryption Standard (DES)* with a 56-bit key and 64-bit block size.
- **Triple DES & Alternative Ciphers (1990s):** Due to DES's short 56-bit key vulnerability to brute-force machines, *Triple DES (3DES)* was adopted. Simultaneously, algorithms like *IDEA* (128-bit key), *Blowfish* (Bruce Schneier), and *RC4* (Rivest) emerged.
- **AES Open Competition (1997–2001):** NIST hosted a multi-year global open evaluation for a replacement cipher. The Rijndael design (by Vincent Rijmen and Joan Daemen) won due to its elegant algebraic SPN structure, efficiency, and resistance to linear/differential cryptanalysis, formally becoming the *Advanced Encryption Standard (AES)*.

Q4 [10 Marks] Classical Substitution Ciphers & Frequency Analysis

- **(a) Caesar Cipher:** Replaces each character with a letter shifted by key k: C = (P + k) \bmod 26. Key space is trivial (25 keys); vulnerable to exhaustive brute force in seconds.
- **(b) Monoalphabetic Cipher:** Maps each alphabet letter to an arbitrary permutation of 26 letters (Key space 26! \approx 4 \times 10^{26}). While brute-force is infeasible, it preserves English letter frequency (e.g., 'E' is 12.7%, 'T' is 9.1%), falling rapidly to *frequency analysis*.
- **(c) Playfair Cipher:** Encrypts digrams (pairs of letters) using a 5×5 key matrix. Rules handle digrams in same row, column, or rectangle. By operating on digrams (25 \times 25 = 625 combinations), it flattens single-letter frequency spikes, though digram frequency analysis still breaks it.
- **(d) Vigenère Cipher:** Polyalphabetic substitution using a repeating keyword to shift successive characters: $C_i = (P_i + K_{i \bmod m}) \bmod 26$. Masks single-letter frequencies. However, Friedrich Kasiski (1863) cracked it by finding distances between repeated ciphertext strings to factor the key length m, reducing it to m parallel Caesar ciphers.

Why frequency analysis succeeds/fails: It succeeds on Caesar and Monoalphabetic ciphers because the statistical distribution of letters is preserved 1:1. It is less effective against Playfair and Vigenère because one plaintext character maps to multiple different ciphertext characters depending on its position.

Q5 [8 Marks] Classical Transposition Techniques, Product Ciphers & The One-Time Pad

- **Rail Fence Cipher:** Writes plaintext diagonally across depth rails and reads row by row. Preserves letter counts but alters character positions.
- **Columnar Transposition:** Plaintext is written in fixed-width rows; columns are read out in order determined by an alphanumeric key word. Easily analyzed using anagramming and vowel-consonant digraph tables.
- **Product Ciphers:** Modern block ciphers combine alternate stages of substitution (confusion) and transposition (diffusion). Neither alone is secure; iterated sequentially in rounds (Shannon's concept), they produce unbreakable cryptosystems.
- **One-Time Pad (OTP):** Plaintext bits are XORed with a truly random, non-repeating key stream of equal length used exactly once: C = P \oplus K. OTP provides *information-theoretic security* (unbreakable even with infinite compute) because every plaintext is equally likely. However, it is **impractical** because key generation requires true physical randomness, and key distribution requires transmitting a key as large as the message itself.

Q6 [12 Marks] Data Encryption Standard (DES) Architecture

DES is a symmetric Feistel block cipher standardized in 1977. Key specifications:

- **Block Size:** 64 bits; **Key Size:** 64 bits input (56 effective bits + 8 parity bits dropped via PC-1).
- **Structure:** Initial Permutation (IP) → 16 Feistel Rounds → 32-bit Half Swap → Inverse Initial Permutation (IP^{-1}).



DES Round Function F: (1) Expansion table (E-box) expands 32-bit R to 48 bits; (2) Key mixing XORs expanded bits with 48-bit subkey K_i; (3) 8 non-linear S-boxes compress 48 bits to 32 bits (6 bits in, 4 bits out; outer bits select row, inner select column); (4) 32-bit Permutation (P-box). **Key Schedule:** 56-bit key split into two 28-bit halves, left-shifted cyclically by 1 or 2 bits per round, compressed via PC-2 to produce sixteen 48-bit subkeys.

Q7 [10 Marks] DES Strength, Weaknesses, 3DES & Meet-in-the-Middle

Strengths: Strong avalanche effect (one bit change impacts 34 bits by round 5); resilient against differential and linear cryptanalysis. **Fatal Weakness:** 56-bit key space (2^{56} \approx 7.2 \times 10^{16}) is too small. In 1998, EFF's "Deep Crack" brute-forced DES in under 56 hours.

Double DES & Meet-in-the-Middle (MitM): Double DES applies $C = E_{K2}(E_{K1}(P))$ expecting 112-bit security. Diffie and Hellman proved it yields only 57-bit effective security: an attacker computes and stores $E_{K1}(P)$ in a hash table (2^{56} steps), then decrypts $D_{K2}(C)$ looking for matches. **Triple DES (3DES):** Uses Encrypt-Decrypt-Encrypt (EDE): $C = E_{K1}(D_{K2}(E_{K3}(P)))$. 3-key 3DES provides 168-bit key length and 112-bit security against MitM.

Q8 [10 Marks] Advanced Encryption Standard (AES) & Comparison with DES

AES operates on a 128-bit block as a 4×4 byte State matrix. Key sizes and rounds: **128-bit key:** 10 rounds; **192-bit:** 12 rounds; **256-bit:** 14 rounds. Each round applies:

- **1. SubBytes:** Non-linear byte substitution using S-box over Galois Field GF(2^8).
- **2. ShiftRows:** Cyclic left shift of rows: row 0 shifts 0 bytes, row 1 shifts 1, row 2 shifts 2, row 3 shifts 3.
- **3. MixColumns:** Multiplies each column by a fixed matrix over GF(2^8) (omitted in final round).
- **4. AddRoundKey:** XORs state matrix with 128-bit round subkey derived via Key Expansion.

AES vs. DES: AES uses a Substitution-Permutation Network (encrypts entire 128-bit block simultaneously), while DES uses Feistel (encrypts half block per round). AES decryption requires separate inverse transformations (InvSubBytes, InvShiftRows, InvMixColumns), whereas DES uses identical hardware reversing subkey order. AES offers vastly superior security (256-bit key) and throughput.